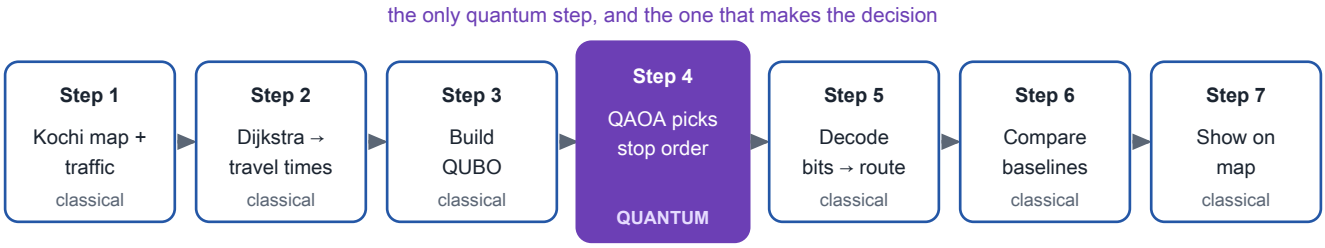
Minnal-Q

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Where and why quantum is used

The short answer: quantum is used in **one place**: choosing the **order in which the ambulance visits patients**. A QAOA circuit in Qiskit encodes every possible visiting order into qubits and is trained so that the safest, fastest route becomes the most likely measurement. Everything around it (map, traffic, checking, display) is classical. **Remove the quantum step and Minnal-Q has no route.**

1. WHERE quantum sits in the pipeline



Steps 1–3 prepare the quantum input · Steps 5–7 read and check the quantum output

Step	What happens	Type	Tool
1	Kochi junctions and hospitals; road times × time-of-day traffic factor	Classical	Python
2	Shortest travel time between every pair of points	Classical	Dijkstra / NetworkX
3	Turn the routing problem into a QUBO cost formula	Classical (prepares quantum input)	NumPy
4	**QAOA circuit finds the best visiting order**	**QUANTUM**	**Qiskit QAOAAnsatz, SparsePauliOp, AerSimulator**
5	Turn measured bitstrings into routes; drop invalid ones	Classical	Python
6	Check against the exact best route and greedy	Classical	Python
7	Draw routes, charts and evidence	Classical	Streamlit + Plotly

2. WHY quantum fits this problem

Reason	Explanation
Routing is a yes/no problem	"Is patient A at stop 1?" is a yes/no question, and a qubit holds exactly one yes/no answer (0 or 1). A full route is a pattern of these answers, so it maps directly onto qubits.
The costs come in pairs	Travel time depends on which two patients are next to each other in the route. Pairwise costs are exactly the form a QUBO captures, and a QUBO is exactly what QAOA is built to solve. There is no forced fit.
The number of routes explodes	The number of possible orders grows as n! (n factorial), as the table below shows. That growth is why routing is a classic benchmark for quantum optimization.
QAOA is built for today's machines	QAOA uses short, shallow circuits and a classical helper. It runs well on simulators and is designed for current noisy quantum computers, including IBM hardware.
It is meaningful, not decorative	The rules say "adding a Hadamard gate or a random quantum circuit does not qualify". In Minnal-Q the circuit is the decision-maker: its output is the route.
It is ready to scale	As quantum hardware grows, the same code can handle more stops. Only the backend changes.

How fast the number of routes grows

Patients (n)	Possible orders (n!)	Qubits needed (n^2)	Where it runs
3	6	9	Live demo on AerSimulator (fast); IBM hardware bonus
4	24	16	AerSimulator experiment
6	720	36	Too big to simulate exactly on a laptop
10	3,628,800	100	Needs future hardware; brute force is already slow
15	about 1.3 trillion	225	Brute force is impractical

Honesty note: at demo size (3–4 patients) a classical computer is still faster. We do **not** claim quantum advantage, and the event page says it is not expected. We show a correct, verified quantum pipeline for a problem that becomes very hard as it grows.

3. HOW the quantum step works

3.1 Every qubit is a yes/no question

Three patients (A = critical, B = medium, C = minor) and three stops need 3 × 3 = **9 qubits**:

	Stop 1	Stop 2	Stop 3
Patient A (critical)	q0	q1	q2
Patient B (medium)	q3	q4	q5
Patient C (minor)	q6	q7	q8

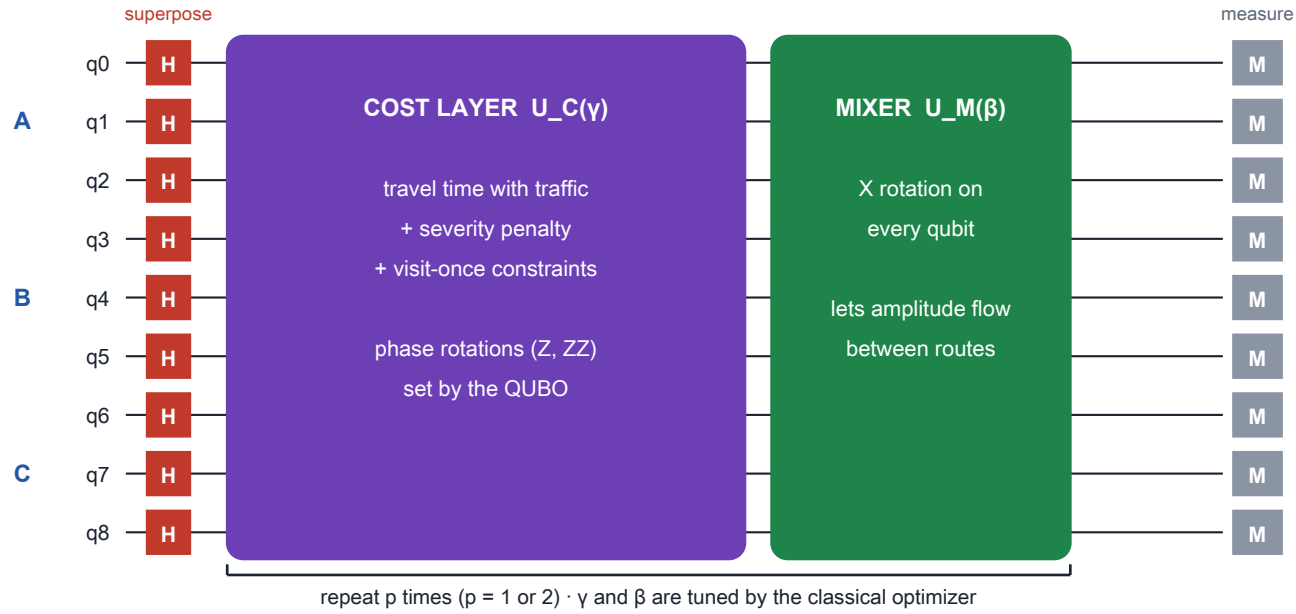
Example: the route **A → B → C** is the bit pattern **100 · 010 · 001** (q0 to q8). Qiskit prints bitstrings in reverse order (q8 first); the decoder handles that.

3.2 The cost formula becomes the circuit's energy

Part of the cost	What it does	Effect
Travel time with traffic	Adds the drive time between consecutive stops at the chosen time of day	Prefers short routes
Severity penalty	Adds cost for visiting a high-severity patient late (weights 1, 2, 4, 8)	Critical patients first
Visit-once constraints	Large cost if a patient is skipped or visited twice	Only valid routes win

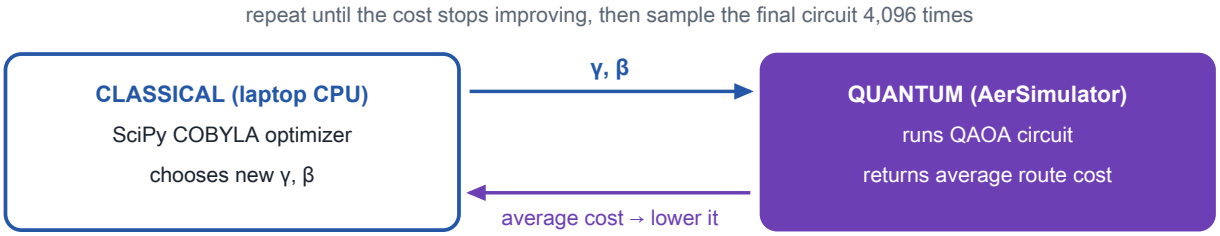
Qiskit converts this formula into a quantum operator (SparsePauliOp). **The best route is the lowest-energy state**, and QAOA searches for it.

3.3 The QAOA circuit



- **H gates (superpose):** start all 9 qubits in an equal mix of every possible bit pattern.
- **Cost layer (γ):** phase rotations that encode every route's cost into the quantum state.
- **Mixer layer (β):** moves probability between routes so that cheaper routes become more likely.
- **Measure:** each run returns one bit pattern, which is one candidate route.

3.4 The hybrid loop (quantum and classical working together)



The classical optimizer never sees the routes directly. It only adjusts the two angles γ and β so that the quantum circuit's average route cost goes down. This is the "hybrid" design the requirements ask teams to explain.

3.5 Measurement gives the answer

Measured bits (q0 to q8)	Decoded route	Valid?	Meaning
100 · 010 · 001	A → B → C	Yes	Critical first: the kind of route Minnal-Q should favour
100 · 001 · 010	A → C → B	Yes	Critical first, different order for the other two
001 · 010 · 100	C → B → A	Yes	Nearest-first style: critical patient last (high cost)
110 · 000 · 001	A twice, B missing	No	Broken route, filtered out by the decoder

Illustrative examples only. Real probabilities are measured on the day from 4,096 shots and shown in the demo's Quantum Lab tab.

4. Qiskit components used

Qiskit component	Role in Minnal-Q
qiskit.quantum_info.SparsePauliOp	Holds the route-cost formula as a quantum operator (Z and ZZ terms)
qiskit.circuit.library.QAOAAnsatz	Builds the QAOA circuit: superposition, cost layer, mixer, p repetitions
qiskit_aer.AerSimulator / EstimatorV2	Simulates the circuit and returns the average cost while training
qiskit_aer SamplerV2	Runs the trained circuit 4,096 times to collect candidate routes
generate_preset_pass_manager	Compiles the circuit for a real device (optional hardware run)
qiskit_ibm_runtime (optional)	Sends the 9-qubit version to an IBM quantum computer; the job ID goes in the README

5. Questions judges may ask

Question	Answer
Is the quantum part meaningful?	Yes. The QAOA circuit makes the routing decision. Without it there is no route; it is not a decorative circuit.
Why not just use brute force?	We do use it, as the answer key that proves QAOA is correct. Brute force breaks down as patients increase (n! growth). QAOA is the approach built to scale with quantum hardware.
Is it faster than classical?	Not at demo size, and we say so openly. The event does not expect quantum advantage. We show a correct, tested pipeline.
Why QAOA and not Grover or VQE?	Our problem is a cost-minimization QUBO. QAOA is designed exactly for that and runs well on today's noisy devices. Grover needs much deeper circuits, and VQE is aimed at chemistry-style problems.
What about real hardware?	Main results come from AerSimulator (allowed by the rules and documented). As a bonus we run the 9-qubit version on IBM hardware and report the job ID, including the effect of noise.
How do you know the answer is right?	Every QAOA route is checked against the exact best route found by brute force. We also test that the QUBO's lowest-energy state equals that exact answer.

6. The 30-second explanation for judges

"Quantum decides the most important thing in Minnal-Q: the order in which the ambulance reaches patients. We turn every possible route into 9 qubits. The QAOA circuit is trained, with a classical optimizer, to make the route that gets the critical patient first, through Kochi traffic, the most likely result. When we measure it, that route comes out on top, and we check it against the exact answer. Minnal means lightning, and that's how fast the right route has to arrive."